

# 9401

## CRC GENERATOR/CHECKER

### FAIRCHILD TTL MACROLOGIC

**DESCRIPTION** – The 9401 Cycle Redundancy Check (CRC) Generator/Checker provides an advanced tool for implementing the most widely used error detection scheme in serial digital data handling systems. A 3-bit control input selects one-of-eight generator polynomials. The list of polynomials includes CRC-16 and CRC-CCITT as well as their reciprocals (reverse polynomials). Automatic right justification is incorporated for polynomials of degree less than 16. Separate clear and preset inputs are provided for floppy disc and other applications. The Error output indicates whether or not a transmission error has occurred. Another control input inhibits feedback during check word transmission. The 9401 is fully compatible with all TTL families.

- GUARANTEED 10 MHz DATA RATE
- EIGHT SELECTABLE POLYNOMIALS
- ERROR INDICATOR
- SEPARATE PRESET AND CLEAR CONTROLS
- AUTOMATIC RIGHT JUSTIFICATION
- FULLY COMPATIBLE WITH ALL TTL LOGIC FAMILIES
- 14-PIN PACKAGE
- TYPICAL APPLICATIONS:

FLOPPY AND OTHER DISC STORAGE SYSTEMS  
DIGITAL CASSETTE AND CARTRIDGE SYSTEMS  
DATA COMMUNICATION SYSTEMS

#### PIN NAMES

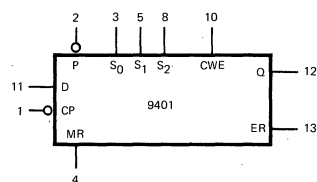
|                  |                                                  |
|------------------|--------------------------------------------------|
| $S_0 - S_2$      | Polynomial Select Inputs                         |
| $\overline{D}$   | Data Input                                       |
| $\overline{CP}$  | Clock (Operates on HIGH-to-LOW Transition) Input |
| $\overline{CWE}$ | Check Word Enable Input                          |
| $\overline{P}$   | Preset (Active LOW) Input                        |
| $\overline{MR}$  | Master Reset (Active HIGH) Input                 |
| $Q$              | Data Output (Note b)                             |
| $ER$             | Error Output (Note b)                            |

| LOADING (Note a) |          |
|------------------|----------|
| HIGH             | LOW      |
| $S_0 - S_2$      | 1.0 U.L. |
| $\overline{D}$   | 1.0 U.L. |
| $\overline{CP}$  | 1.0 U.L. |
| $\overline{CWE}$ | 1.0 U.L. |
| $\overline{P}$   | 1.0 U.L. |
| $\overline{MR}$  | 1.0 U.L. |
| $Q$              | 10 U.L.  |
| $ER$             | 10 U.L.  |

#### NOTES:

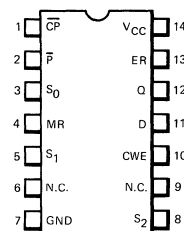
- a. 1 TTL Unit Load (U.L.) = 40  $\mu$ A HIGH/1.6 mA LOW.
- b. The output LOW drive factor is 2.5 U.L. for Military (XM) and 5 U.L. for Commercial (XC) Temperature Ranges.

#### LOGIC SYMBOL



$V_{CC}$  = Pin 14  
GND = Pin 7

#### CONNECTION DIAGRAM DIP (TOP VIEW)

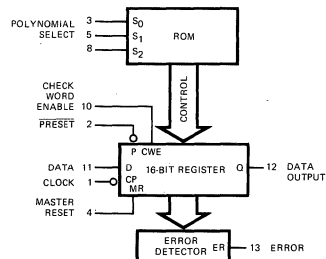


Pins 6 and 9 not connected.

#### NOTE:

The Flatpak version has the same pinouts (Connection Diagram) as the Dual In-line Package.

#### BLOCK DIAGRAM



$V_{CC}$  = Pin 14  
GND = Pin 7

**FUNCTIONAL DESCRIPTION** The 9401 is a 16-bit programmable device which operates on serial data streams and provides a means of detecting transmission errors. Cyclic encoding and decoding schemes for error detection are based on polynomial manipulation in modulo arithmetic. For encoding, the data stream (message polynomial) is divided by a selected polynomial. This division results in a remainder which is appended to the message as check bits. For error checking, the bit stream containing both data and check bits is divided by the same selected polynomial. If there are no detectable errors, this division results in a zero remainder. Although it is possible to choose many generating polynomials of a given degree, standards exist that specify a small number of useful polynomials. The 9401 implements the polynomials listed in *Table 1* by applying the appropriate logic levels to the select pins  $S_0$ ,  $S_1$  and  $S_2$ .

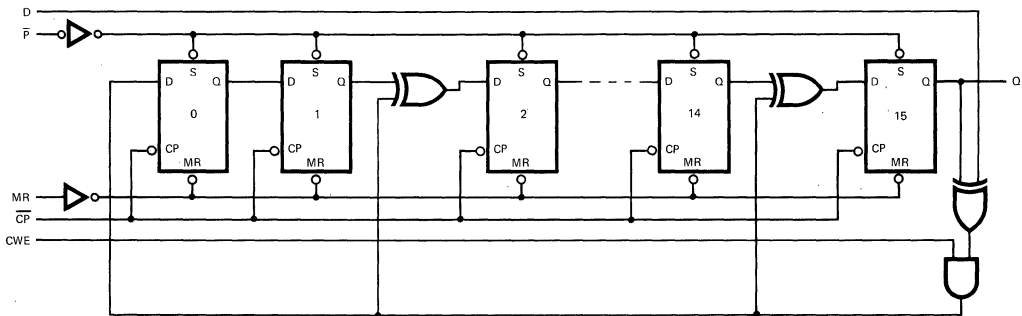
The 9401 consists of a 16-bit register, a Read Only Memory (ROM) and associated control circuitry as shown in the block diagram. The polynomial control code presented at inputs  $S_0$ ,  $S_1$  and  $S_2$  is decoded by the ROM, selecting the desired polynomial by establishing shift mode operation on the register with Exclusive OR gates at appropriate inputs. To generate the check bits, the data stream is entered via the Data inputs (D), using the HIGH-to-LOW transition of the Clock input ( $\overline{CP}$ ). This data is gated with the most significant output (Q) of the register, and controls the Exclusive OR gates (*Figure 1*). The Check Word Enable (CWE) must be held HIGH while the data is being entered. After the last data bit is entered, the CWE is brought LOW and the check bits are shifted out of the register and appended to the data bits using external gating (*Figure 2*).

To check an incoming message for errors, both the data and check bits are entered through the D input with the CWE input held HIGH. The 9401 is not in the data path, but only monitors the message. The Error Output becomes valid after the last check bit has been entered into the 9401 by a HIGH-to-LOW transition of CP. If no detectable errors have occurred during the data transmission, the resultant internal register bits are all LOW and the Error Output (ER) is LOW. If a detectable error has occurred, ER is HIGH.

A HIGH on the Master Reset input (MR) asynchronously clears the register. A LOW on the Preset input ( $\bar{P}$ ) asynchronously sets the entire register if the control code inputs specify a 16-bit polynomial; in the case of 12 or 8-bit check polynomials only the most significant 12 or 8 register bits are set and the remaining bits are cleared.

TABLE 1

| SELECT CODE    |                |                | POLYNOMIAL                             | REMARKS           |
|----------------|----------------|----------------|----------------------------------------|-------------------|
| S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> |                                        |                   |
| L              | L              | L              | $X^{16}+X^{15}+X^{2}+1$                | CRC-16            |
| L              | L              | H              | $X^{16}+X^{14}+X+1$                    | CRC-16 REVERSE    |
| L              | H              | L              | $X^{16}+X^{15}+X^{13}+X^7+X^4+X^2+X+1$ |                   |
| L              | H              | H              | $X^{12}+X^{11}+X^3+X^2+X+1$            | CRC-12            |
| H              | L              | L              | $X^8+X^7+X^5+X^4+X+1$                  |                   |
| H              | L              | H              | $X^8+1$                                | LRC-8             |
| H              | H              | L              | $X^{16}+X^{12}+X^5+1$                  | CRC-CCITT         |
| H              | H              | H              | $X^{16}+X^{11}+X^4+1$                  | CRC-CCITT REVERSE |



**Fig. 1**  
**EQUIVALENT CIRCUIT FOR  $X^{16}+X^{15}+X^2+1$**

DC CHARACTERISTICS OVER OPERATING TEMPERATURE RANGE (unless otherwise noted)

| SYMBOL   | PARAMETER                    |         | LIMITS |       |       | UNITS         | TEST CONDITIONS (Note 1)                              |
|----------|------------------------------|---------|--------|-------|-------|---------------|-------------------------------------------------------|
|          |                              |         | MIN    | TYP   | MAX   |               |                                                       |
| $V_{IH}$ | Input HIGH Voltage           |         | 2.0    |       |       | V             | Guaranteed Input HIGH Voltage                         |
| $V_{IL}$ | Input LOW Voltage            | XM      |        |       | 0.7   | V             | Guaranteed Input LOW Voltage                          |
|          |                              | XC      |        |       | 0.8   |               |                                                       |
| $V_{CD}$ | Input Clamp Diode Voltage    |         |        | -0.9  | -1.5  | V             | $V_{CC} = \text{MIN}, I_{IN} = -18 \text{ mA}$        |
| $V_{OH}$ | Output HIGH Voltage          | XM      | 2.4    | 3.4   |       | V             | $V_{CC} = \text{MIN}, I_{OH} = -400 \mu\text{A}$      |
|          |                              | XC      | 2.4    | 3.4   |       |               |                                                       |
| $V_{OL}$ | Output LOW Voltage           | XM & XC |        | 0.35  | 0.4   | V             | $V_{CC} = \text{MIN}, I_{OL} = 4.0 \text{ mA}$        |
|          |                              | XC      |        | 0.45  | 0.5   | V             | $V_{CC} = \text{MIN}, I_{OL} = 8.0 \text{ mA}$        |
| $I_{IH}$ | Input HIGH Current           |         |        | 1.0   | 40    | $\mu\text{A}$ | $V_{CC} = \text{MAX}, V_{IN} = 2.7 \text{ V}$         |
|          |                              |         |        |       | 1.0   | mA            | $V_{CC} = \text{MAX}, V_{IN} = 5.5 \text{ V}$         |
| $I_{IL}$ | Input LOW Current            |         |        | -0.22 | -0.36 | mA            | $V_{CC} = \text{MAX}, V_{IN} = 0.4 \text{ V}$         |
| $I_{OS}$ | Output Short Circuit Current |         | -15    |       | -100  | mA            | $V_{CC} = \text{MAX}, V_{OUT} = 0 \text{ V}$ (Note 3) |
| $I_{CC}$ | Supply Current               |         |        | 70    | 110   | mA            | $V_{CC} = \text{MAX}$ , Inputs Open                   |

AC CHARACTERISTICS:  $V_{CC} = 5.0 \text{ V}$ ,  $T_A = 25^\circ\text{C}$

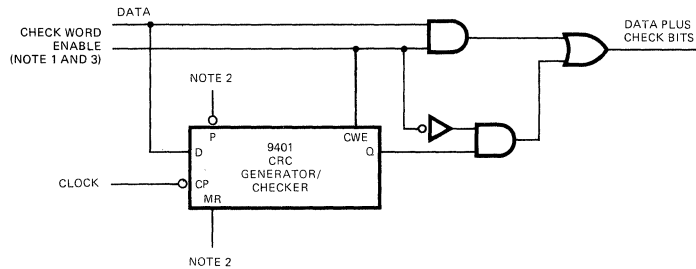
| SYMBOL           | PARAMETER                                                 | LIMITS |                 |     | UNITS | CONDITIONS   |                       |
|------------------|-----------------------------------------------------------|--------|-----------------|-----|-------|--------------|-----------------------|
|                  |                                                           | MIN    | TYP<br>(Note 2) | MAX |       |              |                       |
| $f_{\text{max}}$ | Maximum Clock Frequency                                   | 10     | 18              |     | MHz   | Fig. 3, 4, 5 | $C_L = 15 \text{ pF}$ |
| $t_{\text{PHL}}$ | Propagation Delay, Clock, MR to Data Output               |        | 30              | 55  | ns    |              |                       |
| $t_{\text{PLH}}$ |                                                           |        | 40              | 60  | ns    |              |                       |
| $t_{\text{PHL}}$ | Propagation Delay, Preset to Data Output                  |        | 40              | 60  | ns    |              |                       |
| $t_{\text{PLH}}$ |                                                           |        | 40              | 60  | ns    |              |                       |
| $t_{\text{PHL}}$ | Propagation Delay, Clock,<br>MR or Preset to Error Output |        | 40              | 60  | ns    |              |                       |
| $t_{\text{PLH}}$ |                                                           |        | 40              | 60  | ns    |              |                       |

AC SET-UP REQUIREMENTS:  $V_{CC} = 5.0 \text{ V}$ ,  $T_A = 25^\circ\text{C}$

| SYMBOL        | PARAMETER                             | LIMITS |     |     | UNITS | CONDITIONS |                       |
|---------------|---------------------------------------|--------|-----|-----|-------|------------|-----------------------|
|               |                                       | MIN    | TYP | MAX |       |            |                       |
| $t_{wCP} (L)$ | Clock Pulse Width (LOW)               | 35     |     |     | ns    | Fig. 2     | $C_L = 15 \text{ pF}$ |
| $t_{sD}$      | Set-up Time, Data to Clock            | 55     | 35  |     | ns    | Fig. 6     |                       |
| $t_{sCWE}$    | Set-up Time, CWE to Clock             | 55     | 35  |     | ns    |            |                       |
| $t_h$         | Hold Time, Data and CWE to Clock      | 0      | −10 |     | ns    |            |                       |
| $t_{wP} (L)$  | Preset Pulse Width (LOW)              | 40     | 30  |     | ns    | Fig. 4     |                       |
| $t_{wMR} (H)$ | Master Reset Pulse Width (HIGH)       | 35     | 25  |     | ns    | Fig. 6     |                       |
| $t_{rec}$     | Recovery Time, MR and Preset to Clock | 50     | 25  |     | ns    | Fig. 4, 5  |                       |

NOTES:

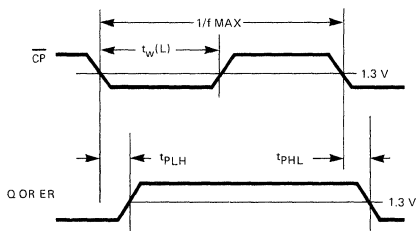
- For conditions shown as MIN or MAX, use the appropriate value specified under recommended operating conditions for the applicable device type.
- Typical limits are at  $V_{CC} = 5.0 \text{ V}$ ,  $T_A = 25^\circ\text{C}$ .
- Not more than one output should be shorted at a time.



NOTES:

1. Check word Enable is HIGH while data is being clocked, LOW during transmission of check bits.
2. 9401 must be reset or preset before each computation.
3. CRC check bits are generated and appended to data bits.

**Fig. 2**  
**CHECK WORD GENERATION**



**Fig. 3**  
**PROPAGATION DELAYS,**  
 **$\overline{\text{CP}}$  TO Q AND  $\overline{\text{CP}}$  TO ER**

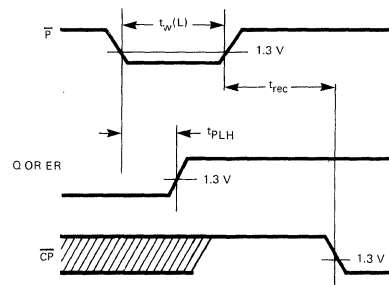


Fig. 4  
PROPAGATION DELAYS,  $\bar{P}$  TO Q AND ER  
PLUS RECOVERY TIME  $\bar{P}$  TO  $\bar{CP}$

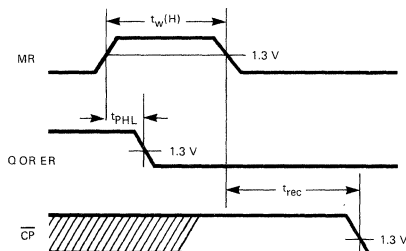
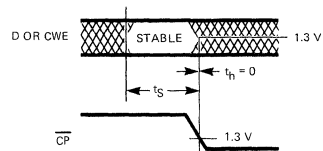


Fig. 5  
PROPAGATION DELAYS, MR TO Q AND ER  
PLUS RECOVERY TIME, MR TO  $\overline{CP}$



**Fig. 6**  
**SET-UP AND HOLD TIMES,**  
**D TO  $\overline{\text{CP}}$  AND CWE TO  $\overline{\text{CP}}$**